

Method Evaluation: Fuzzy Exercise Assistant

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Introduction and Objectives

In this evaluation paper, we will evaluate the fuzzy exercise assistant FCM demo's first version. We had lastly presented our code and discussed some improvements we could make and some factors we had overlooked. First, we will give a short summary on our method, indicate the factors that had to be resolved and optimistically give our solutions to these matters.

Our Demo

We ran an FCM

Inputs

The system accepts four inputs from the user. Each input is converted to a decimal value between 0 and 1, where 0 represents the lowest value and 1 represents the highest. In the interface, the user rates themselves on a scale of 1 to 10 for each input, and this value is automatically normalized to the $[0, 1]$ range before being transmitted to the FCM

Input	Type	Description	Scale Meaning
BMI	Numeric (normalised to 0–1)	User enters their BMI value; the interface converts it to a 0–1 scale.	1 = very high BMI 0 = very low BMI
Fitness_Lvl	Scale: 0–1	How experienced the user is with exercise, from complete beginner to professional level.	0 = no experience 1 = professional / expert
Goal_Muscle Gain	Scale: 0–1	How strongly the user wants to gain muscle mass as a goal.	0 = not a goal 1 = primary goal
Goal_WeightLoss	Scale: 0–1	How strongly the user wants to lose weight as a goal.	0 = not a goal 1 = primary goal

Input Details

BMI is not the only input that is entirely user-reported as a preference or goal; it is a physical measurement. The user enters their BMI value directly, and the system maps it to the [0, 1] range. A higher normalized BMI value means the system will steer away from high-impact exercises and toward lower-intensity options.

Fitness Level indicates how experienced the user is with exercise. A value of 0 indicates that the user has no fitness experience, while a value of 1 represents someone who trains at a high or even professional level. This input has a strong influence on whether HIIT and advanced strength programs are appropriate.

“Muscle Gain” and **“Weight Loss”** are treated as independent goals. A user can set both of these goals at the same time (for example, they may want to build muscle while losing fat), choose just one, or choose neither. This independence is intentional, rather than forcing the user to prioritize one goal over the other, FCM manages the combined effect of these goals on the results through the weight matrix.

Outputs

The system generates recommendations for four different types of exercises. Each output is a value between 0 and 1; here, 1 indicates that the exercise type in question is strongly recommended for this user profile, while 0 indicates it is irrelevant. The outputs are not mutually exclusive; a user can receive high scores for multiple types at the same time, which means the system is recommending a combination.

Output	What it means	When it is most relevant
Rec_Light_Cardio	Low-intensity cardiovascular exercise such as walking, light cycling, or swimming at a relaxed pace.	Particularly relevant for users with high BMI or those new to exercise, where joint stress must be minimised.
Rec_Strength	Resistance and weight training exercises aimed at building muscle and increasing metabolic rate.	Central recommendation for users with a muscle gain goal; also supports weight loss through increased resting metabolism.
Rec_HIIT	High-Intensity Interval Training — short bursts of intense effort alternated with rest periods.	Most suitable for users with a good fitness base. Not recommended for high BMI or complete beginners due to injury and cardiovascular risk.
Rec_Beginner	General beginner programs using bodyweight movements and foundational exercise patterns.	Relevant for users with little or no exercise experience regardless of their specific goals. Does not correlate with muscle gain or weight loss goals directly.

Interpreting the Output

Output values should be interpreted as a relative ranking rather than a strict rule. The exercise type with the highest score is the most suitable starting point for the user. Types scoring above approximately 0.6 can be considered actively recommended; those scoring below 0.3 can be considered irrelevant to the user's current profile.

An important design decision is that Rec_Beginner is not tied to goals such as muscle gain or weight loss. This is intentional: a beginner program is determined solely by the user's experience level, not their goals. A complete beginner aiming to build muscle should still start with basic exercises before moving on to specialized strength training.

The results will be displayed on the interface as horizontal bars ranging from 0 to 1, allowing users without technical knowledge to easily see at a glance which types of exercises the system recommends most. Where appropriate, the interface will also include links to sample workouts for each recommended type.

Limitations

After completing the coding, we identified several factors that required reconsideration. The weights in our demo code were assigned intuitively, which to some may not make sense or may lack a solid basis. Therefore, we needed to find valid arguments supporting the correlations between our input and output nodes.

The second issue we had to consider was the interface and how it would be presented to the user, especially a beginner exerciser. The most difficult issue we face, and one we are still struggling to address, is the evaluation method. How do we know if the system works properly? What if the weights are actually incorrect? What happens if the user has a chronic disease and should not exercise at all? My colleague and I have reviewed various articles to find reasonable arguments and have proposed different solutions to these problems, but further research is still required.

Basing our Input Weights

The issue with the weights concerned the reliability of the correlations between inputs and outputs. We needed to base our node weights on actual research rather than solely on our own intuition. To establish our weights, we reviewed several papers and attempted to ground our weights in this research while still incorporating our professional intuition. For example, a user with a muscle gain goal was assigned a very high weight for strength training.

It should be noted that beginner-level workouts have no significant correlation with either muscle gain goals or weight loss goals.

1. BMI

Benisek, A. (n.d.). *Safe Workouts When You're Overweight*. WebMD.
<https://www.webmd.com/obesity/safe-workouts-when-you-are-overweight>

- This source suggests that individuals who are overweight should engage in lighter cardio activities such as yoga or cycling, and states: "With some movements, you may have more pain in your joints with obesity because you're carrying more weight. For this reason, it's important to find movements that aren't harsh on your body."

troutman, C. (2017, May 15). *Obesity and Exercise*. Obesity Medicine Association. <https://obesitymedicine.org/blog/obesity-and-exercise/>

"While this anaerobic exercise is not always appreciated for its role in treating obesity, studies show that it can increase resting metabolic rate and reduce metabolic conditions like type 2 diabetes and heart disease. The American College of Sports Medicine recommends at least two sessions of resistance training weekly. Strength training can help prevent muscle loss during weight loss, while stronger muscles can help protect joints and improve posture."

- Although there are differing perspectives on muscle-strengthening exercises, we chose not to assign them a negative correlation but rather a low positive correlation.

2. FITNESS LEVEL

The fitness level rows were filled intuitively.

3. GOAL OF MUSCLE GAIN

Hickson, R. C. (1980). Interference of strength development by simultaneously training for strength and endurance. *European Journal of Applied Physiology and Occupational Physiology*, 45(2-3), 255–263. <https://doi.org/10.1007/bf00421333><https://doi.org/10.1007/bf00421333>

- According to this paper and many others, cardiovascular exercise does slow muscle growth when performed concurrently with strength training. Light cardio can still be done. However, because the variables are treated independently, it was rated negatively.

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"HIIT promotes increases in lean mass, maximal strength, early RFD, and lower limb muscle power in healthy older and young individuals."

- Therefore, HIIT was rated positively.

4. GOAL OF LOSING WEIGHT

Laguaite, M. (2023, February 25). *The Best Exercises to Lose Weight*. WebMD. <https://www.webmd.com/fitness-exercise/ss/slideshow-exercises-weightloss>

- Both cardio and HIIT aid in weight loss, but HIIT is more effective. Strength training also contributes to weight loss and helps maintain it after weight loss has been achieved.

Input Node	Rec_Light_Cardio	Rec_Strength	Rec_HIIT	Rec_Beginner
BMI	+0.7	+0.2	-0.3	+0.8
Fitness_Lvl	-0.2	+0.7	+0.9	-0.8
Goal_MuscleGain	-0.2	+1.0	+0.2	0.0
Goal_WeightLoss	+0.6	+0.3	+0.8	0.0

The Interface

We will build a simple HTML page where users can interact with the assistant and rate themselves from 1 to 10 on the following input areas: BMI, fitness level, muscle gain goal, and weight loss goal. These ratings will then be converted to decimal values within the range [0, 1]. We will also include a pop-up warning for individuals who may have chronic diseases or injuries, advising them to consult their doctor beforehand, as we are not qualified to provide medical advice in this domain. The interface will be as simple as possible, potentially incorporating fitness-themed visuals for added motivation. After the output is generated, we will also provide links to sample workouts corresponding to the recommended exercise types.

Evaluation Method

This is the most challenging aspect of our work, as it is difficult to objectively evaluate whether an exercise is good for a specific individual. Instead, we must assess whether the user feels the recommendations are appropriate for them. We considered several options:

Expert evaluation: Sending the recommendations to fitness coaches who could evaluate the results, provide feedback for improvement, and suggest weight adjustments.

User evaluation: Having users actually follow the recommendations for at least one week and then evaluate them via a questionnaire.

The second option seemed more feasible in our case, as we do not have access to a large number of fitness coaches, but we can reach users directly.

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

The questionnaire would include the following statements, rated from Strongly Disagree to Strongly Agree:

- The recommended exercises matched my personal fitness goals.
- The recommendations were appropriate for my current fitness level.
- The recommendations felt personalized to me (not generic).
- The variety of exercises recommended was adequate.
- I would have chosen similar exercises on my own.
- The recommendations were useful for planning my workouts.
- I trust that the exercise recommendations are safe for me.
- The recommendations made logical sense given my inputs.
- I would feel comfortable following these recommendations without checking with a professional first.
- The assistant seemed to understand my needs.
- I would recommend this assistant to a friend with similar fitness goals.

- I intend to use this assistant again in the future.
- I plan to follow the exercise recommendations this week.
- This assistant would help me stay consistent with exercise.
- I would prefer using this assistant over searching for exercises on my own.
- The recommendations avoided exercises that might be risky for me.
- I felt the assistant appropriately considered my physical limitations.
- The recommended exercises were not carrying a risk for me.

After collecting the results from all users, we will generate summaries and include open-ended questions for further improvement. However, the evaluation of this assistant remains ambiguous because, unless a user breaks a bone or sustains a serious injury, any form of exercise will provide some benefit albeit to varying degrees. There is still some improvement needed.